DECEMBER 31, 2022 · AMHERST, NOVA SCOTIA

Allison Holthoff, 37.

A mother of three. She came to the emergency room with severe abdominal pain.

She was triaged at 11:14 a.m. Then she sat down to wait.

She got worse in the waiting room for hours. She died that day.

No one was positioned to see it happening.

This is about a gap in the system, not about any person. The family's legal claims have not been tested in court. Sources: CBC News, CTV News Atlantic, Global News, 2023.

NOT ONLY ALLISON

673 people died in Nova Scotia emergency departments in one year.

673

Up from **406** three years earlier.*

Waits are long, and people get worse while they wait: **1 in 3** emergency patients in one study deteriorated **within 4 hours**.

After triage, nobody is watching the waiting room.

*2023-24 vs 2020-21. Counts all deaths after triage and before discharge or admission, not only waiting-room deaths. Nova Scotia Health also cites sicker patients.
Sources: Nova Scotia Health figures reported by CTV News and The Globe and Mail, July 2024. Reed et al., SNAP40-ED (250 patients, 32.8%), Emergency Care Journal, 2021.

OUR SOLUTION

ALLEY watches while you wait.

1 · HEALTH RECORD

Who they are

Why they came in, triage level, history. **Higher risk, lower bar to alert.**

2 · WEARABLE BAND

Their vitals

Heart rate and blood oxygen, **every second**, not once every few hours.

3 · CAMERA

Their posture

Upright, slumped, or lying down. **No facial recognition. No video stored.**

Check now

When the signs agree, the nurse is told **who to check first, and why.** ALLEY does not replace the nurse.

WORKING TODAY

Live camera posture detection, the cross-check, the nurse dashboard, the AI alert and handoff.

SIMULATED TODAY

Wearable data (from a control panel) and patient records (mock data).

WHERE THE AI IS

AI sees, cross-checks, and explains.

1 · AI SEES

Computer vision

A vision model reads body posture from the live camera and notices a **slump or a collapse**.

2 · ALLEY CROSS-CHECKS

Vitals + posture + history

It tracks the trend in the vitals and weighs it against posture and the health record. **Two signals agree: Check now.**

3 · AI EXPLAINS

Plain words for the nurse

Claude writes the alert in one clear sentence, and the **handoff summary** of everything that happened in the wait.

Clear rules a clinician can set, so every alert states its reason. The nurse always decides.

Posture: OpenCV with a pose-estimation model. Priority is set by explainable rules; the language model writes text only and never changes who is seen first. ALLEY never tells anyone they are fine.

THE IMPACT

Earlier warning. More time to act.

65 min

earlier. In an Edinburgh ER study, a wearable caught deterioration a median **65 minutes before** manual checks.*

Already in N.S.

Dartmouth General began piloting wireless pulse and oxygen monitors on waiting patients in Jan 2024.

Next step

A silent pilot in one Nova Scotia ER. We measure how early ALLEY would have flagged each patient.

IN HONOUR OF

Allison Holthoff, and the 673. So that no one waits unwatched.

*Reed et al., SNAP40-ED, Emergency Care Journal, 2021 (n=31 of 250 patients; a research armband, not ALLEY). ALLEY has not yet been tested with patients. We are not promising a number of lives saved; we are promising to measure. Dartmouth: Nova Scotia Health, Sept 2024.